Lesson for the course "Human-Computer Interaction"

Design Principles and Heuristics

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Design principles

- Generalizable abstractions intended to orient designers towards thinking about different aspects of their design
- They are concerned with
  - **What to provide and what not to provide at the interface level**
  - **What users should see/hear/touch/smell/taste and do when carrying out their tasks using an interactive product**
- Design principles are derived from a mix of theory-based knowledge, experience and common-sense

Design principles

- Several principles have been proposed
- Here we see:
  - **Visibility**
  - **Feedback**
  - **Constraints**
  - **Consistency**
  - **Affordance**
  - **Simplicity**
- You may find lots of examples online about when design principles are not applied…
  - https://www.plerdy.com/blog/20-sajtov-s-uzhasnym-juzabiliti/
  - https://www.mockplus.com/blog/post/bad-web-design

Visibility

- Visibility is concerned with making relevant parts of the design visible, i.e., **making the tasks at hand easy to see and find**
- All the functional parts must be visible and must provide a clear message about what the user can do through them

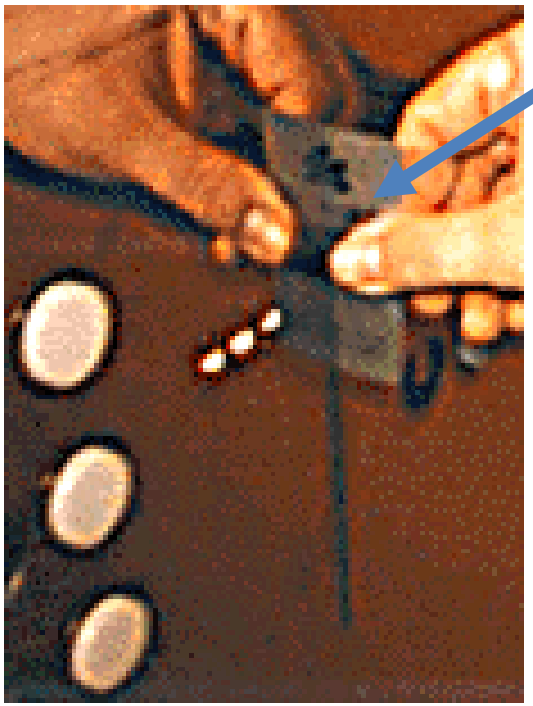
Visibility

- The more visible functions are, the more likely it is that users will be able to know what to do next
- In contrast, when functions are out of sight, it makes them more difficult to find and to know how to use
- Example



- This is a control panel for an elevator
  - How does it work?
  - Push a button for the floor you want?
  - Nothing happens. Push any other button? Still nothing. What do you need to do?
  - It is not visible what to do!

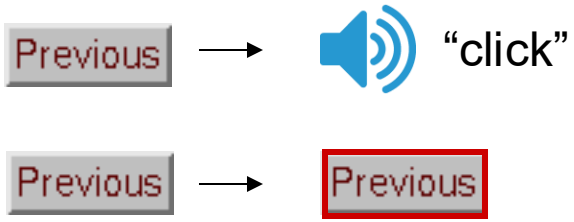
Visibility



- …you need to insert your room card in the slot by the buttons to get the elevator to work!
- How would you make this action more visible?
  - Make the card reader more obvious
  - E.g., provide an auditory message, that says what to do (problem: in which language?)
  - E.g., provide a big label next to the card reader that flashes when someone enters (or if someone presses the buttons without having first used the card)

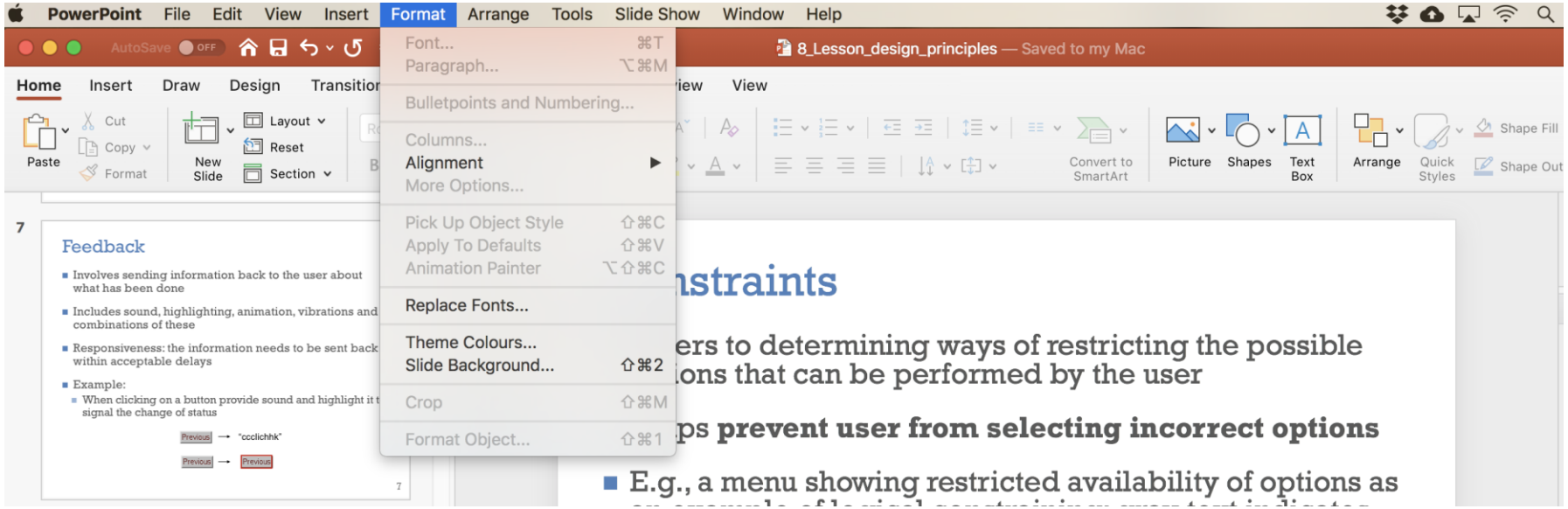
Feedback

- Involves sending **information back to the user about what has been done**
- Includes sound, highlighting, animation, vibrations and combinations of these
- **Responsiveness**: the information needs to be sent back within acceptable delays
- Example:
  - When clicking on a button, provide sound and highlight it to signal the change of status

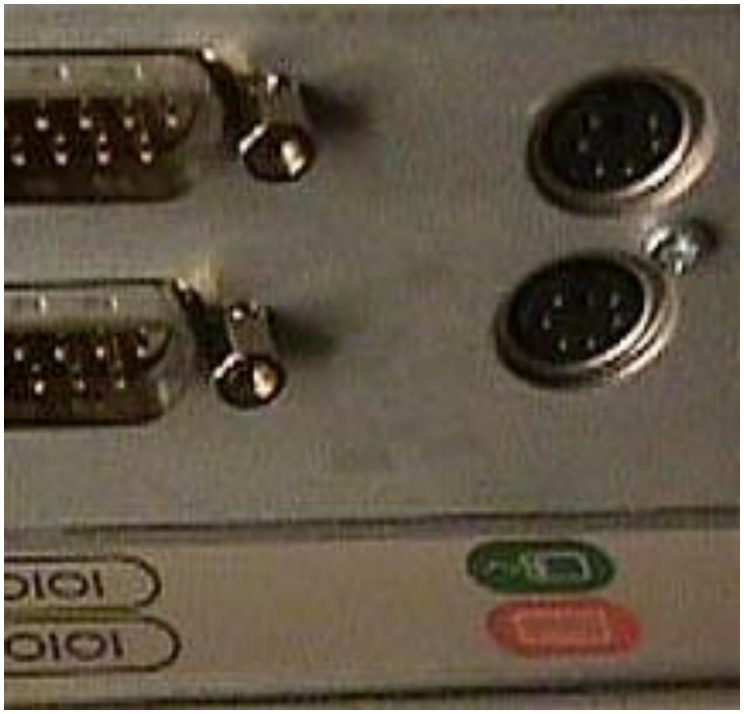


Constraints

- Refers to determining ways of **restricting the possible actions that can be performed by the user**
- Helps **prevent user from selecting incorrect options**
- E.g., a menu showing restricted availability of options as an example of logical constraining: gray text indicates deactivated options



Constraints: ambiguous design



- Where do you plug the mouse?
- Where do you plug the keyboard?
  - Top or bottom connector?
  - Do the color-coded icons help?

How to design them more logically



- A.: provides direct adjacent mapping between icon and connector

- B.: provides color coding to associate the connectors with the labels



Consistency

- Refers to design interfaces to have **similar operations and use similar elements for achieving similar tasks**
- For example: shortcuts
  - **Always** use ctrl key plus first initial of the command for an operation:
    - *ctrl+C* : copy
    - *ctrl+P* : print
    - *ctrl+S* : save
    - *ctrl+O* : open
    - *ctrl+N* : new
- The main benefit of consistent interfaces is that they are **easier to learn and use**
- Consistency at a higher level: between app and website, or between all software of a company (e.g., Microsoft Office suite)

Affordance

- Refers to an attribute of an object that **allows people to know how to use it** (i.e., possibilities for action)
  - E.g., a mouse button **invites** pushing, a door handle affords pulling



- Norman (1988) used the term to discuss the design of everyday objects
  - When the affordances of a physical object are perceptually obvious it is easy to know how to interact with it

Affordance

- Screen-based interfaces are virtual and do not have affordances like physical objects
- Norman argues it does not make sense to talk about interfaces in terms of "real" affordances
- Instead, screen-based interfaces are better conceptualized as "perceived" affordances
  - Are essentially **learned conventions** of arbitrary **mappings** between action and effect at the interface
  - E.g., scrollbars afford moving up and down, icons afford clicking on
  - Some mappings are better than others

Simplicity

- At usability level
  - **Simple things work better and are easier to use** (strong correlation with usability)
- At UX level
  - **Simple things are typically more beautiful** (strong correlation between simplicity and aesthetics)
- *"KISS"* ("Keep It Simple Stupid")
- *"Less is more"*



Design factors for simplicity

- **Visual clutter**
- **Number of colors**
- **Symmetry**
- **Figure-ground contrast**

Example of bad design:



- In a few weeks we will learn about the Gestalt laws of visual perception

10 usability heuristics for UI design

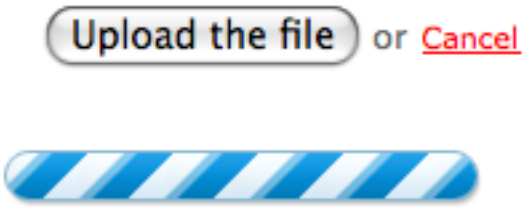
- Jakob Nielsen's 10 general principles for interaction design
- They are called "*heuristics*" because they are broad rules of thumb and not specific usability guidelines
- They are used mainly as the basis **for evaluating systems**
  - They provide a framework for **heuristic evaluation**

10 usability heuristics for UI design

- **Visibility of system status**
- **Match between system and the real world**
- **User control and freedom**
- **Consistency and standards**
- **Error prevention**
- **Recognition rather than recall**
- **Flexibility and efficiency of use**
- **Aesthetic and minimalist design**
- **Help users recognize, diagnose, and recover from errors**
- **Help and documentation**

Heuristic: Visibility of system status

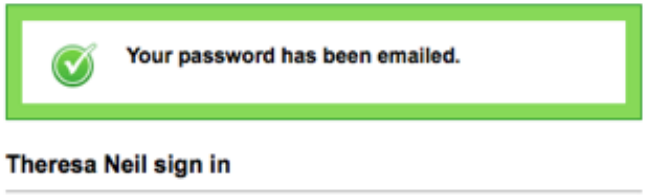
- The system should always keep users informed about what is going on, through appropriate feedback within reasonable time
- The user must always know what the status of the system is

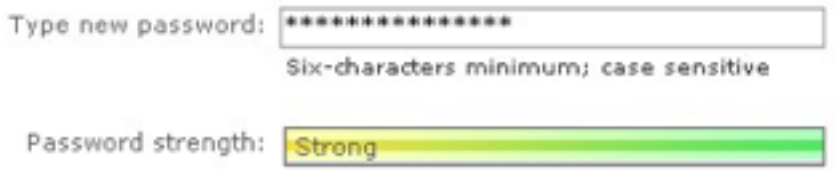


- E.g., progress indicators communicate that the system is working, and reduce uncertainty — preventing users from, e.g., tapping the same button multiple times because they weren't sure if the first time worked

Heuristic: Visibility of system status

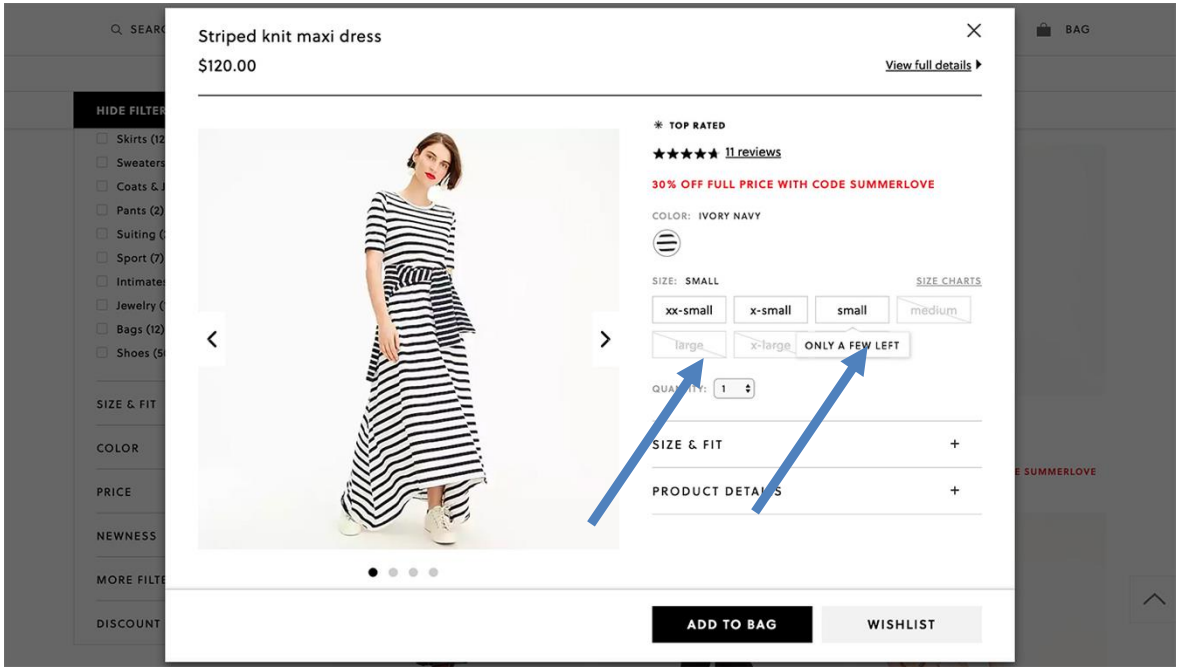
- **A lack of information often equates to a lack of control**
- Whenever users interact with a system, they need to know whether the interaction was successful
- Provide appropriate feedback for a user action serves to keep users informed of the current status and to allow them to steer the interaction in the right direction, without wasting effort





Heuristic: Visibility of system status

- You don't need always to show **all** the states of a system, especially the internal status, usually not needed by users
- Yet sometimes it is useful to users, e.g., inventory size

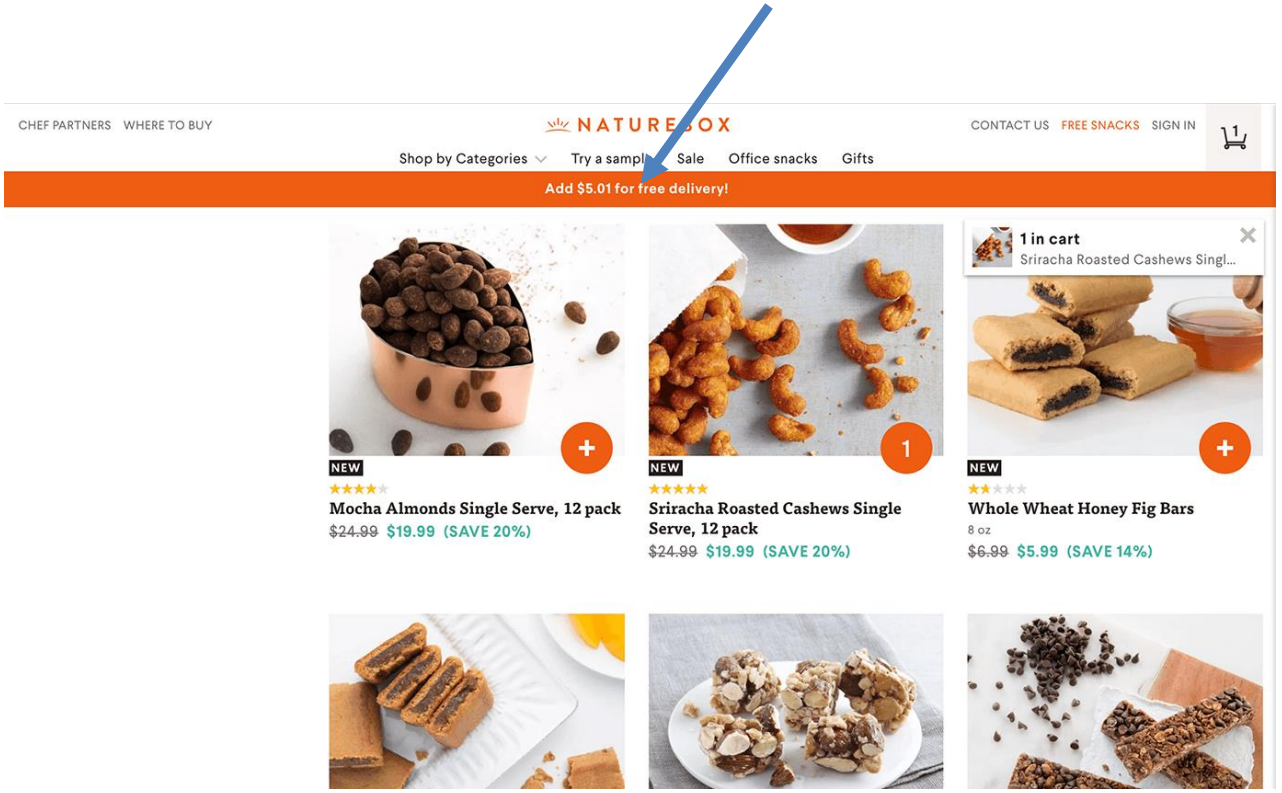


The J.Crew site displays a notification that there are "Only a Few Left!" when the user moves the mouse over low-stock sizes of a product. Some sizes are already sold out, and thus shown crossed out in light grey.

- If people know that there are just a few items left, they are more likely to act immediately

Heuristic: Visibility of system status

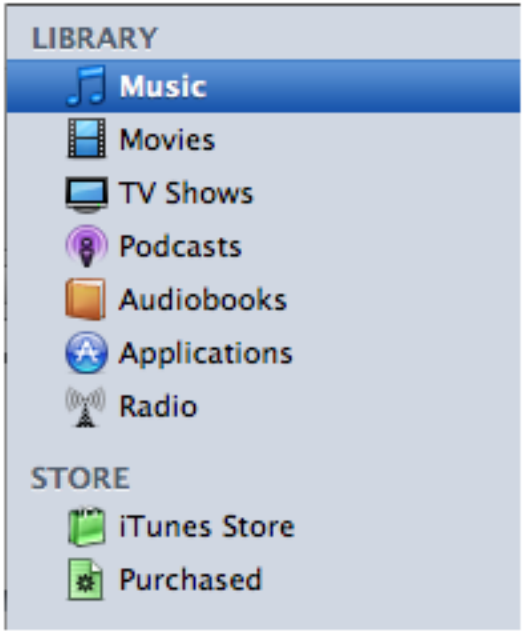
- Communicating how far away a user currently is from qualifying for free shipping or another type of deal can also encourage additional purchases



NatureBox.com: A banner across the top of the page communicates how much more money the user needs to spend to qualify for free shipping

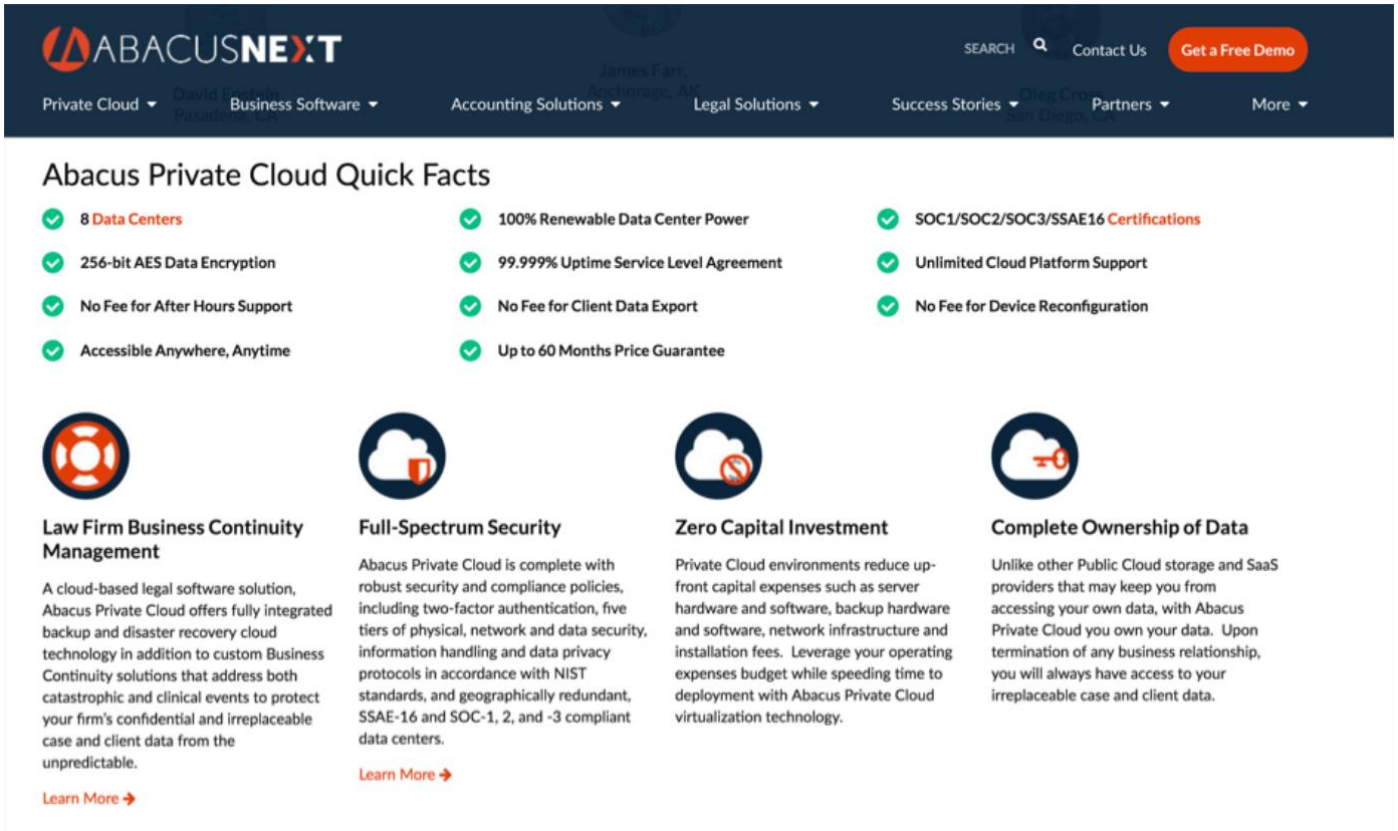
Heuristic: Match between system and the real world

- The system should speak the users' language, with words, phrases and concepts **familiar** to the user, rather than system-oriented terms
- Follow real-world conventions, making information appear in a natural and logical order





Heuristic: Match between system and the real world





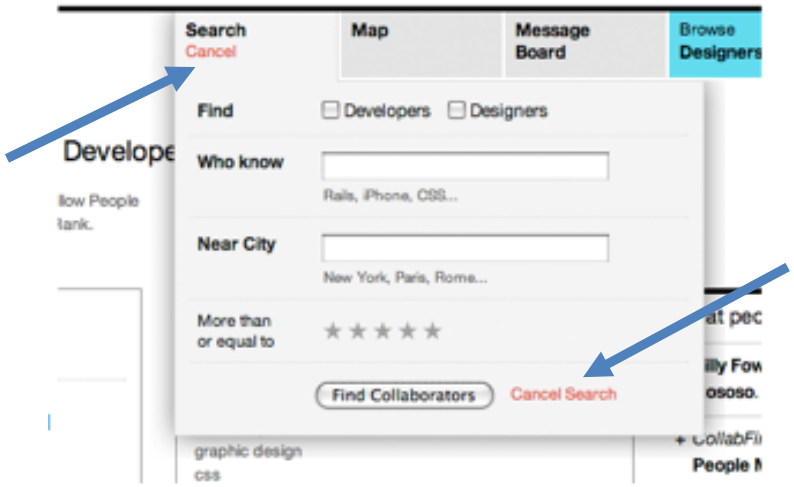
The About page for Abacus legal software's website uses obscure, overly technical language that places much burden on the user to connect the words with their meaning. It is unnecessarily complex and uses obscure phrases that only a specialized IT professional (not a lawyer or common user) would understand

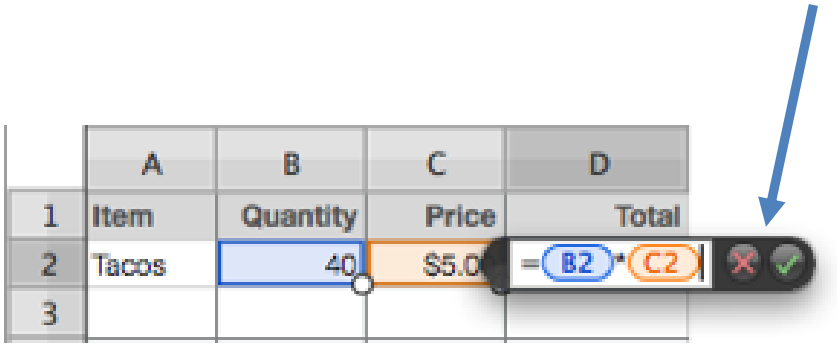
Heuristic: Match between system and the real world

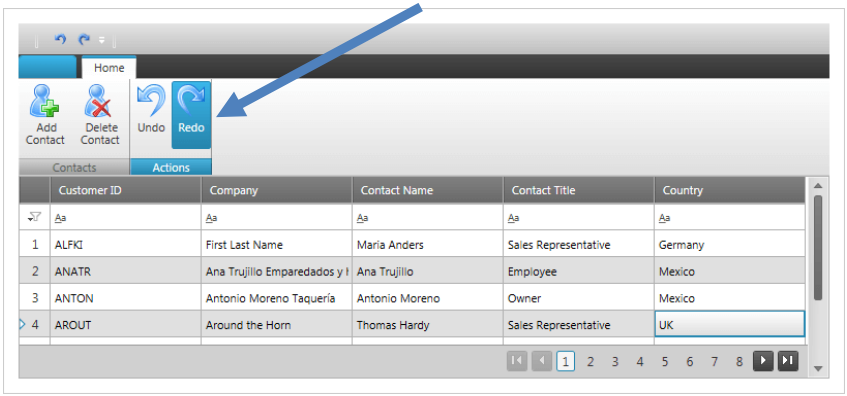
- Interfaces that follow real-world conventions and make information appear in a natural and logical order demonstrate empathy and acknowledgement for users
- UX designers should **never assume that their own interpretations and understanding of words or objects match those of their users**
- If people don't understand the terms used on a site, not only will they feel unsure and ignored, but many will be forced to go elsewhere to find explanations or even to complete their task
- Familiar words are also better for **SEO** (Search Engine Optimization), because they are what users search for

Heuristic: User control and freedom

- Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue
- Support **undo** and **redo**







Heuristic: Consistency and standards

- Users should not have to wonder whether different words, situations, or actions mean the same thing
- Follow platform **conventions**



Word, Excel, and PowerPoint all use the ***same style toolbar*** *with the same primary menu options: Home, Insert, Page Layout… Consistency results in efficiency and perceived intuitiveness*

Heuristic: Error prevention

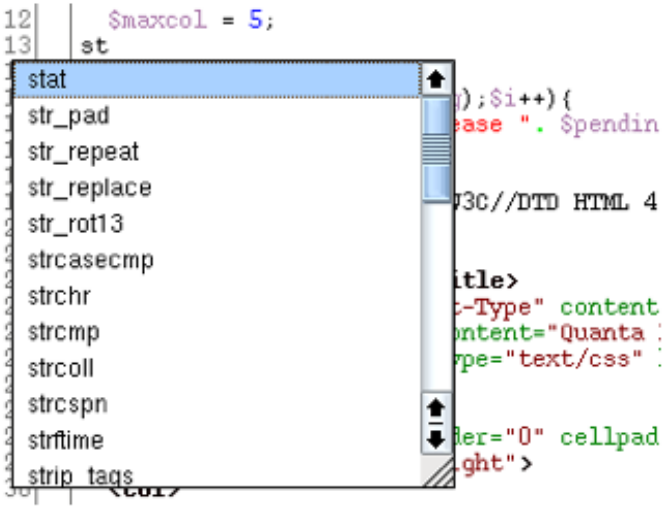
- Even better than good error messages is a careful design which **prevents** a problem from occurring in the first place
- Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action



Disable the update button after it is clicked, so the user cannot update the post twice by accident

Heuristic: Recognition rather than recall

- Minimize the user's memory load by making objects, actions, and options visible
- **The user should not have to remember information from one part of the interface to another**
- Instructions for use of the system should be visible or easily retrievable whenever appropriate



Type ahead for coding in a development environment

Heuristic: Flexibility and efficiency of use

- **Accelerators** — unseen by the novice user — may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users
- Allow users to tailor frequent actions (e.g., shortcuts)



Common Shortcuts

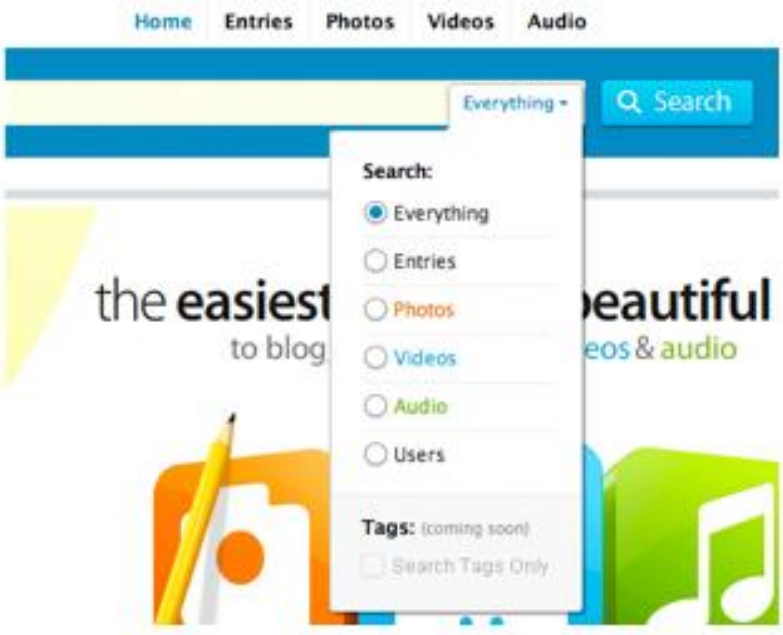
Add Action	**Return**
New Window	**⌘N**
Synchronize with Server	**^⌘S**
Clean Up	**⌘K**
Planning Mode	**⌘1**
Context Mode	**⌘2**
Inbox	**⌥⌘1**
Quick Entry	**^⌥Space**

Quick Entry's shortcut can be customized in Preferences

https://youtu.be/LoTdRTBB8BQ

Heuristic: Aesthetic and minimalist design

- **Interfaces should not contain information that is irrelevant or rarely needed**
- Every extra unit of information in an interface **competes** with the relevant units of information and diminishes their relative visibility
- https://youtu.be/ZgbRmeWDgd0



Aesthetics, *example:*

Contrast: bold text is used for the two labels in the search (i.e., search and tags)

Repetition: the orange, blue, and green text match the color of the media types

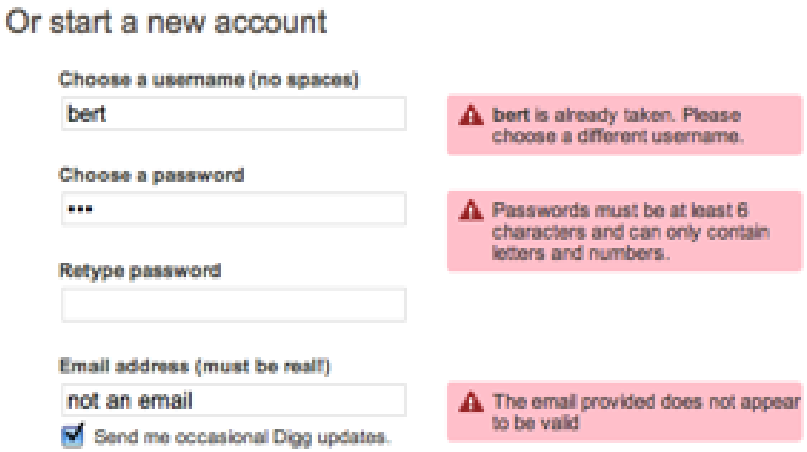
Alignment : strong left alignment of text, right aligned drop down

Proximity: a light rule is used to separate tags from the other options

Heuristic: Help users recognize, diagnose, and recover from errors

- Error messages should be expressed in plain language (no codes), **precisely indicate the problem, and constructively suggest a solution**

Good design example:



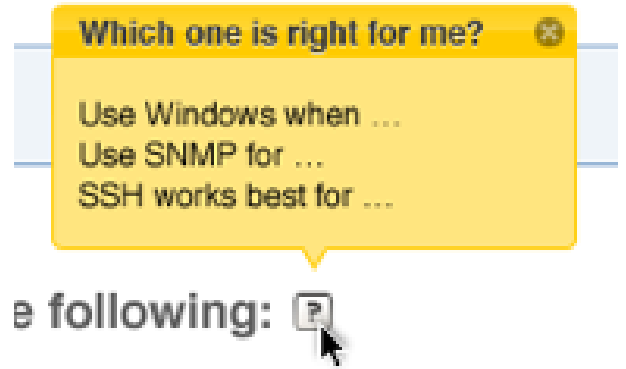
Heuristic: Help and documentation

- Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation
- Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large

Help video

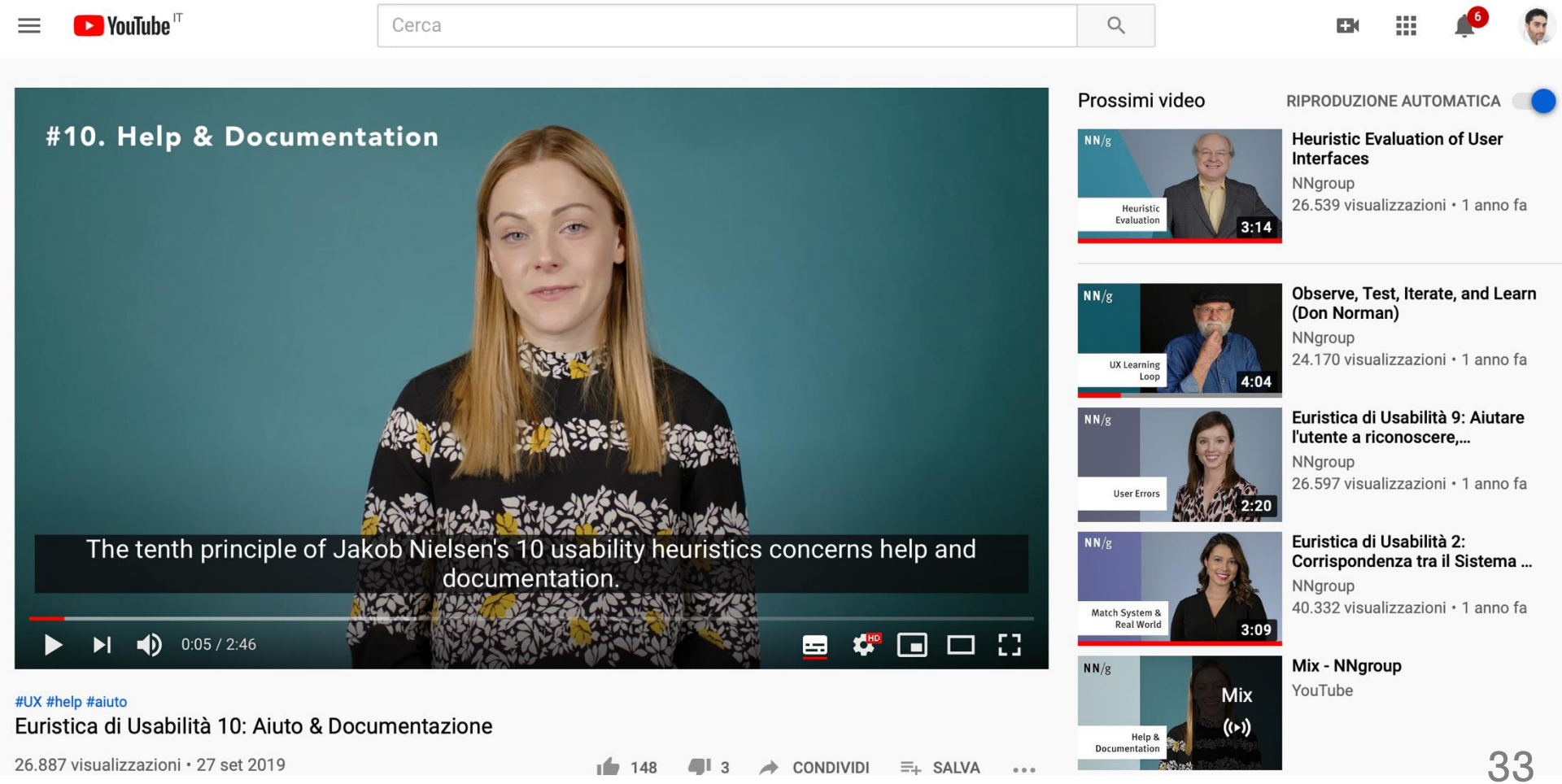


Help tip



Heuristic: Help and documentation

- https://youtu.be/iIQVRzatb50



Evaluation using the heuristics

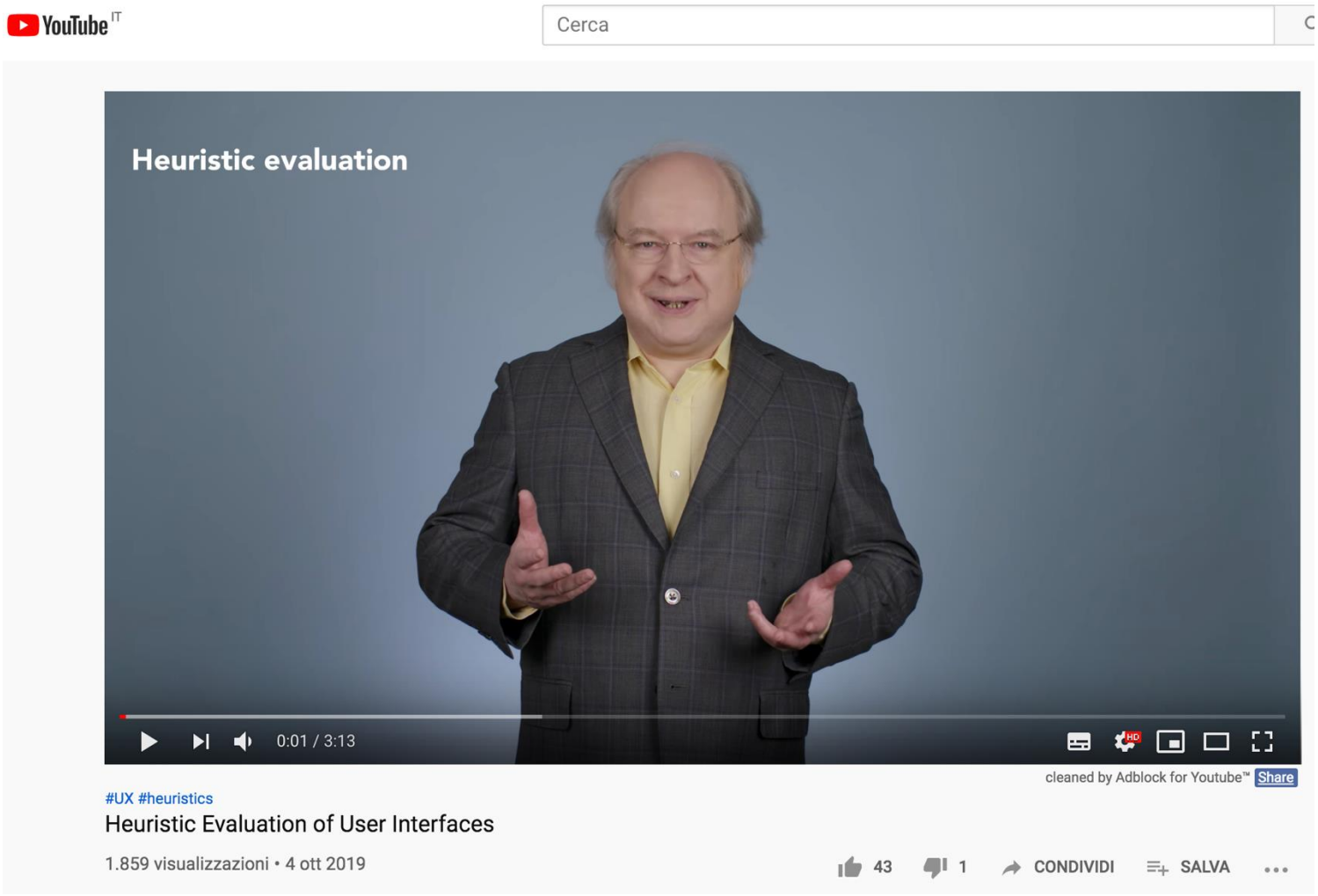
- Designers and researchers evaluate aspects of the interface against the appropriate heuristics
- Those doing the heuristic evaluation **go through the interface several times**, inspecting the various interaction elements and comparing them with the list of usability heuristics
- During each iteration, usability problems will be identified and ways of fixing them may be suggested
- NOTE: **the heuristic evaluation does not substitute the tests with users!**

Severity Ratings for Usability Problems

- Nielsen also proposes to **rate the usability problems in order of importance**

0	I don't agree that this is a usability problem at all
1	Cosmetic problem only. Need not be fixed unless extra time is available on project
2	Minor usability problem: fixing this should be given low priority
3	Major usability problem: important to fix, so should be given high priority
4	Usability catastrophe: imperitave to fix this before product can be released

The words of the expert



- https://youtu.be/6Bw0n6Jvwxk

Universal design

- Universal design is the design of buildings, products or environments **to make them accessible to all people, regardless of age, disability or other factors**

- Accessibility issues can creep in throughout the project life cycle
  - The most **expensive** accessibility issues often come **after** the completion of a project
- In countries with strong accessibility legislations, companies can find themselves in costly lawsuits

Universal design

- In general, companies and public institutions are accountable to provide equal access to *all* users
- Besides legal matters, accessibility can benefit your users and also improve the brand of your product
- So it is important that you plan for and focus your efforts to design for accessibility in the first place

The 7 principles of universal design

- The Center for Universal Design at North Carolina State University (1997) proposed the following 7 principles:

1. **Equitable Use**
2. **Flexibility in Use**
3. **Simple and Intuitive Use**
4. **Perceptible Information**
5. **Tolerance for Error**
6. **Low Physical Effort**
7. **Size and Space for Approach and Use**

- Each principle above is succinctly defined and contains a few brief guidelines that can be applied to design processes in any realm: physical or digital

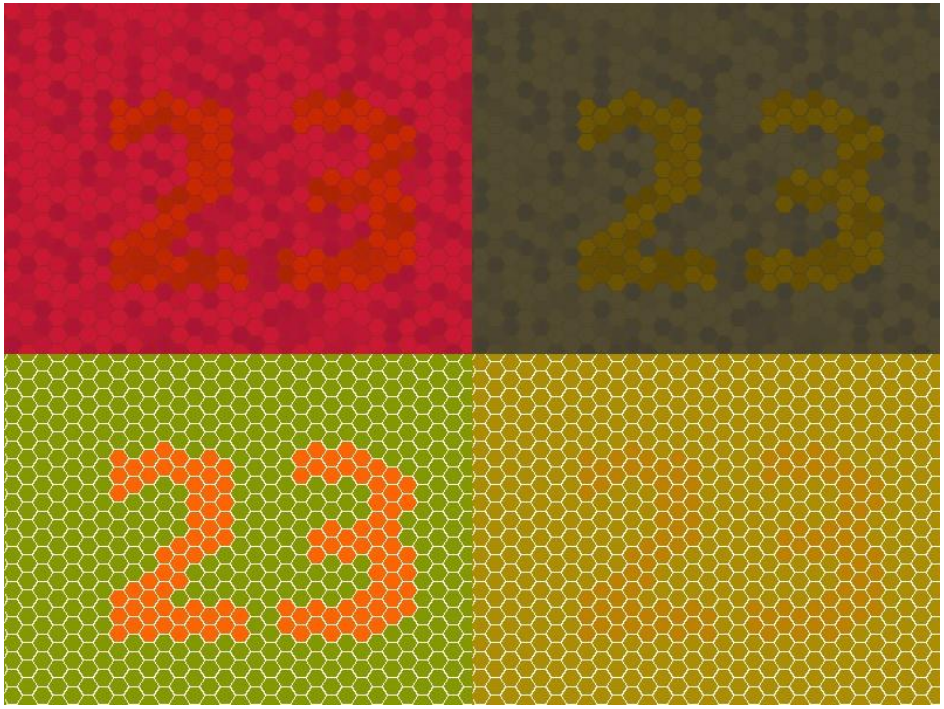
Principle 1: Equitable Use

- **The design is useful and marketable to people with diverse abilities**

- **Guidelines**
  - 1a. Provide the same means of use for all users: identical whenever possible; equivalent when not
  - 1b. Avoid segregating or stigmatizing any users
  - 1c. Provisions for privacy, security, and safety should be equally available to all users
  - 1d. Make the design appealing to all users

Principle 1: Equitable Use – Example

- Design example: **Use strong colour contrast to avoid stigmatizing users with colour blindness**
- Colour blindness affects approximately 1 in 12 men (8%) and 1 in 200 women (0.5%) in the world



Colour blindness - deuteranomaly (red/green distinction). On the left are two pictures as seen by a person with "normal" vision. On the right, the same pictures are simulated as seen by a person with deuteranomaly.

Principle 1: Equitable Use – Example

- When you choose colours for your design, **make sure to avoid red/green combinations**
- You can avoid segregating or stigmatizing your users by designing colour palettes with **strong contrast**

Principle 2: Flexibility in Use

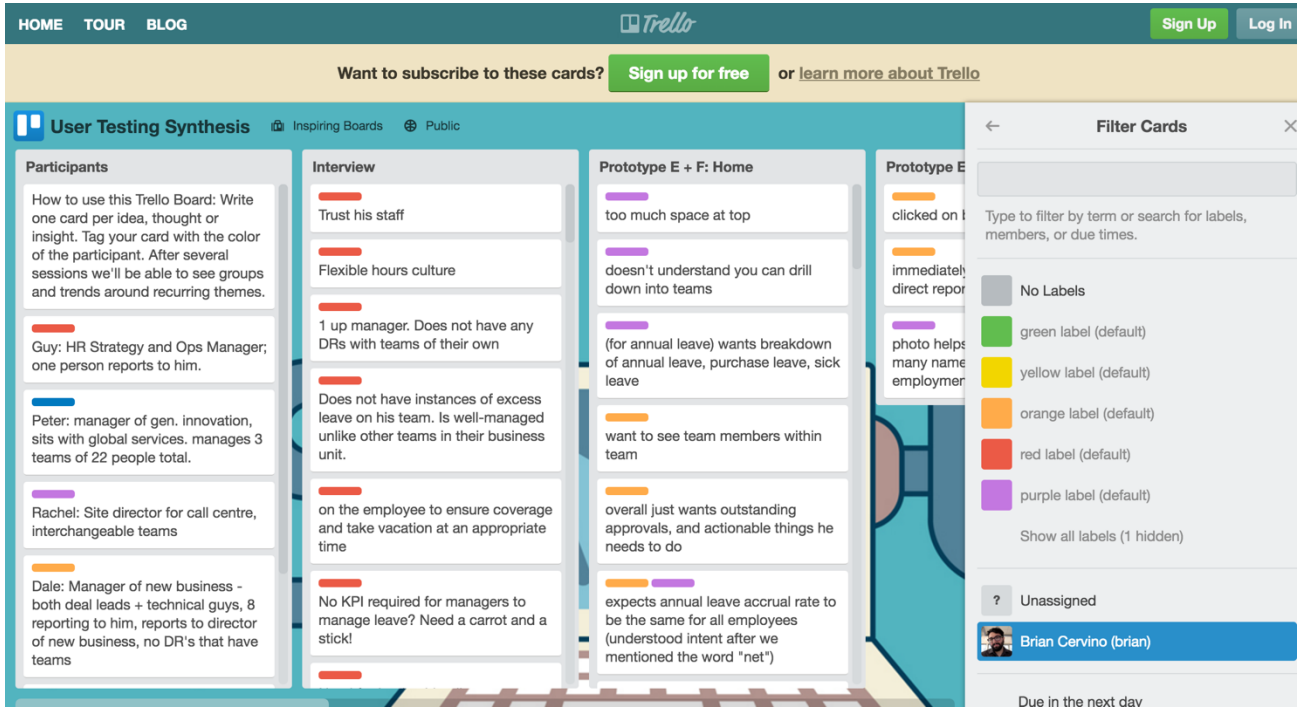
- **The design accommodates a <u>wide range of individual preferences and abilities</u>**

- **Guidelines**
  - 2a. Provide choice in methods of use
  - 2b. Accommodate right- or left-handed access and use
  - 2c. Facilitate the user's accuracy and precision
  - 2d. Provide adaptability to the user's pace

Principle 2: Flexibility in Use – Example

- Design example: **Provide customization for dashboards**
- Customization is a technique to accommodate a wide range of individual preferences and abilities
- It enables users to choose and organize what they see on a website and how they will use it
- Dashboards are good examples for customization
  - Many enterprise systems and project management applications have customizable dashboards (e.g., trello.com)
  - Depending on their various work tasks and needs, users can select what they want to see on the dashboard and how they want to use it

Principle 2: Flexibility in Use – Example



Trello is a web-based project management application. It uses boards, lists and cards to help users to organize and prioritize their projects with flexibility

It offers a range of customizable dashboard options; here, the user can choose the colour of the cards to display on a board

Principle 3: Simple and Intuitive Use

- **The use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level**

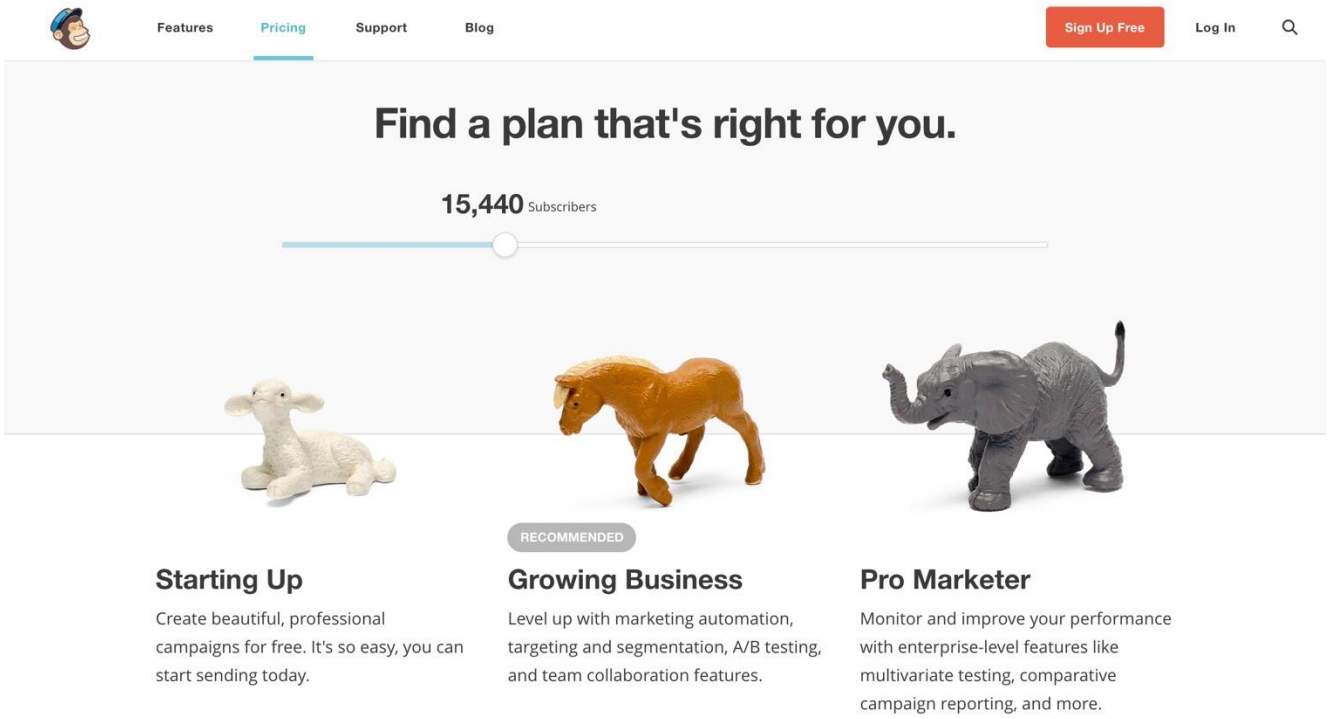
- **Guidelines**
  - 3a. Eliminate unnecessary complexity
  - 3b. Be consistent with user expectations and intuition
  - 3c. Accommodate a wide range of literacy and language skills
  - 3d. Arrange information consistent with its importance
  - 3e. Provide effective prompting and feedback during and after task completion

Principle 3: Simple and Intuitive Use – Example

- Design example: **Reduce visual clutter with Progressive Disclosure**

- Progressive Disclosure is an IxD technique
  - By disclosing information *progressively*, interaction designers reveal only the essentials, and help users manage the complexity of feature-rich websites or applications
  - It reduces visual clutter and removes irrelevant information on the screen
  - It prioritizes information to display on the screen based on user needs and interactions
  - On top of that, it allows the user to drill in for more information, usually without a page load
  - Progressive Disclosure reduces users' cognitive loads and helps them focus on the tasks at hand

Principle 3: Simple and Intuitive Use – Example



MailChimp is a web-based email marketing service. On its pricing page, it uses progressive disclosure to display service plans. A user can interact with the slider to communicate the number of subscribers to whom she will be reaching out using MailChimp. Based on the input, MailChimp displays a subset of three service plans on the screen and recommends one of them to the user

Principle 4: Perceptible Information

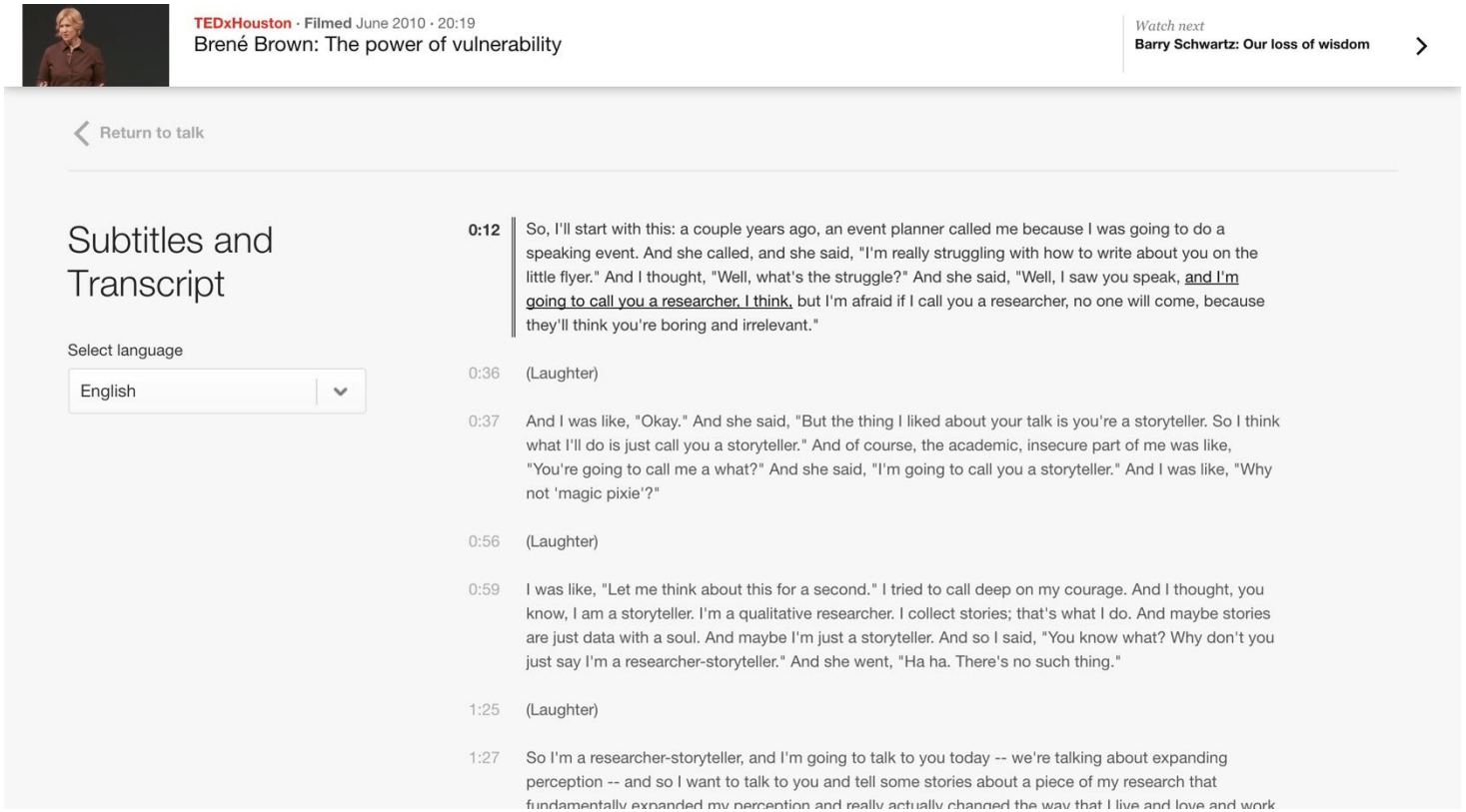
- **The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities**

- **Guidelines**
  - 4a. Use different modes (pictorial, verbal, tactile) for redundant presentation of essential information
  - 4b. Provide adequate contrast between essential information and its surroundings
  - 4c. Maximize "legibility" of essential information
  - 4d. Differentiate elements in ways that can be described (i.e., make it easy to give instructions or directions)
  - 4e. Provide compatibility with a variety of techniques or devices used by people with sensory limitations

Principle 4: Perceptible Information – Example

- Design example: **Enable users with hearing impairments to watch videos with Video Transcription**
- Multimedia requires multisensory experiences, especially videos
- People with hearing problems or language barriers have difficulty consuming information from videos
- You can remove these barriers by providing video transcriptions and/or subtitles
  - Video transcriptions and subtitles add an alternative channel
  - They allow users to consume words and information from the video without relying on hearing
- Designing with accessibility in mind can be a win-win situation for both the company *and* the users

Principle 4: Perceptible Information – Example



TED's website provides subtitles and transcriptions for their online videos. It also provides the option to select a language. That's a huge service for the hearing-impaired user, and it's a great service for everybody else who prefers to read instead of viewing and hearing videos. In turn, it's also good for TED as they reach more users and increase their popularity

Principle 5: Tolerance for Error

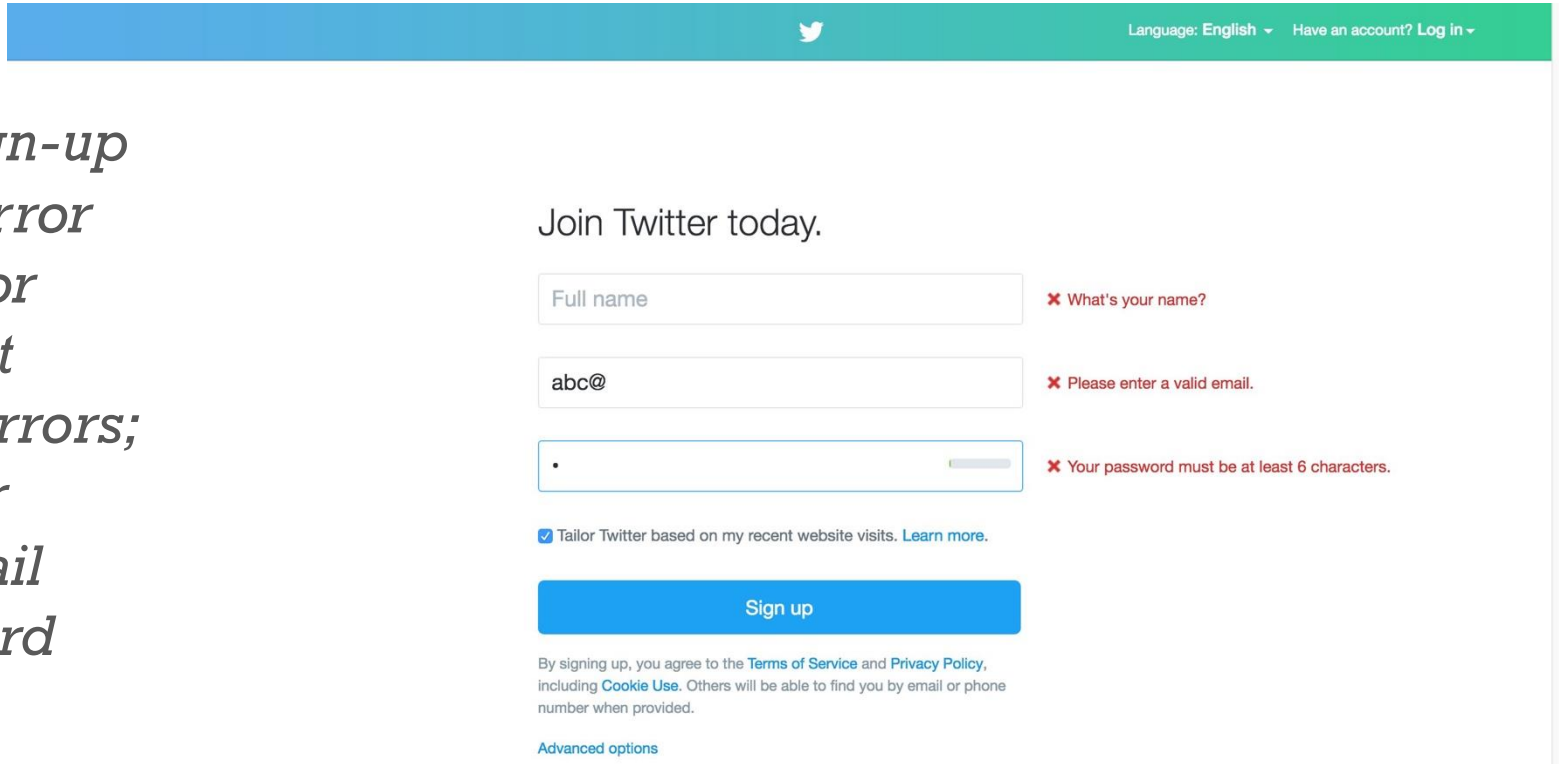
- **The design minimizes hazards and the adverse consequences of accidental or unintended actions**

- **Guidelines**
  - 5a. Arrange elements to minimize hazards and errors: most used elements, most accessible; hazardous elements eliminated, isolated, or shielded
  - 5b. Provide warnings of hazards and errors
  - 5c. Provide fail safe features
  - 5d. Discourage unconscious action in tasks that require vigilance

Principle 5: Tolerance for Error– Example

- Design Example: **Reduce submission error with Form Validation**
- Form Validation minimizes and prevents user errors

Twitter's sign-up form with error messages for input format validation errors; it checks for correct email and password format



Principle 6: Low Physical Effort

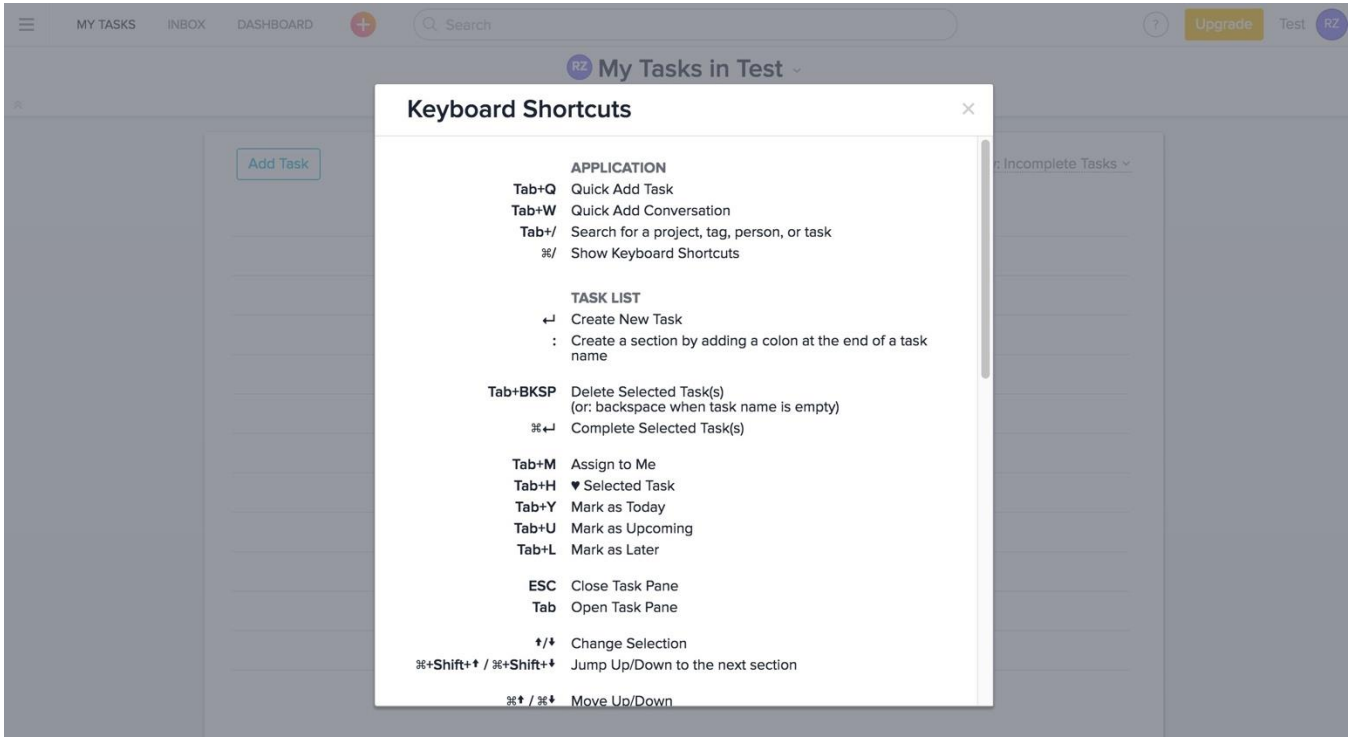
- **The design can be used efficiently and comfortably and with a minimum of fatigue**

- **Guidelines**
  - 6a. Allow user to maintain a neutral body position
  - 6b. Use reasonable operating forces
  - 6c. Minimize repetitive actions
  - 6d. Minimize sustained physical effort

Principle 6: Low Physical Effort– Example

- Design Example: **Minimalize mouse usage with Keyboard Shortcuts**
- Keyboard shortcuts reduce the need to move from the keyboard to the mouse for simple tasks
- Most browsers such as Chrome, Firefox, and Safari provide keyboard shortcuts to perform tasks such as copy (Ctrl + C) and paste (Ctrl + V)
- You can design superb interactive experiences by adding appropriate keyboard shortcuts
- They will improve the navigation and make websites easier to use for all users

Principle 6: Low Physical Effort– Example



Asana is a web-based project management platform. It allows teams to track and manage work items. Asana has a rich set of keyboard shortcuts. These shortcuts reduce mouse usage and minimize repetitive actions.

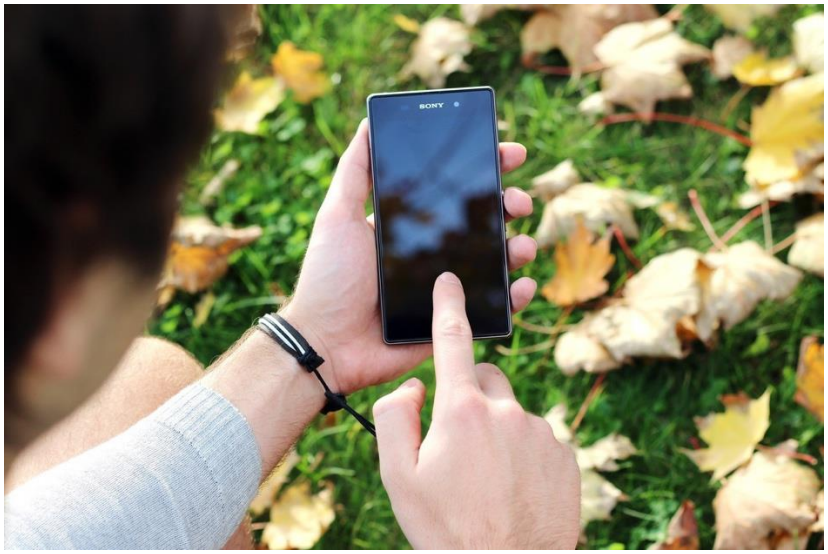
Principle 7: Size and Space for Approach and Use

- **Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility**

- **Guidelines**
  - 7a. Provide a clear line of sight to important elements for any seated or standing user
  - 7b. Make reach to all components comfortable for any seated or standing user
  - 7c. Accommodate variations in hand and grip size
  - 7d. Provide adequate space for the use of assistive devices or personal assistance

Principle 7: Size and Space for Approach and Use – Example

- Design example: **Consider the Target Area of your website when it is on mobile devices**
- A small target area can be a problem on mobile devices because it is more difficult to select with precision
- So, **make sure your touch target areas take the human physical factors into consideration**



Exercise

- **Analyze** a bad website example chosen from https://colorlib.com/wp/bad-websites/ (bad websites in year 2025):
  - Through the lenses of the **Nielsen's 10 heuristics**
  - Through the lenses of **the 7 principles of Universal Design**

Homework

- Select an app and a website **and analyze them through the lenses of the Nielsen's 10 heuristics** as well as the Budd's heuristics
- Select an app and a website and **analyze them through the lenses of the 7 principles of Universal Design**
- Suggested readings:
  - https://www.nngroup.com/articles/visibility-system-status/
  - https://www.nngroup.com/articles/match-system-real-world/
  - https://www.nngroup.com/articles/false-consensus/

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